How-To Library

Informatica

Implementing Data Engineering Integration with Google Dataproc

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Abstract

You can integrate Informatica® Data Engineering Integration with Google Dataproc to run mappings and workflows in a Google cloud Hadoop implementation. This article describes how to perform pre-implementation tasks to integrate with the Dataproc cluster, configure the domain and tools, and access Google cloud sources.

Supported Versions

- Data Engineering Integration 10.4, 10.5.1.1

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Overview

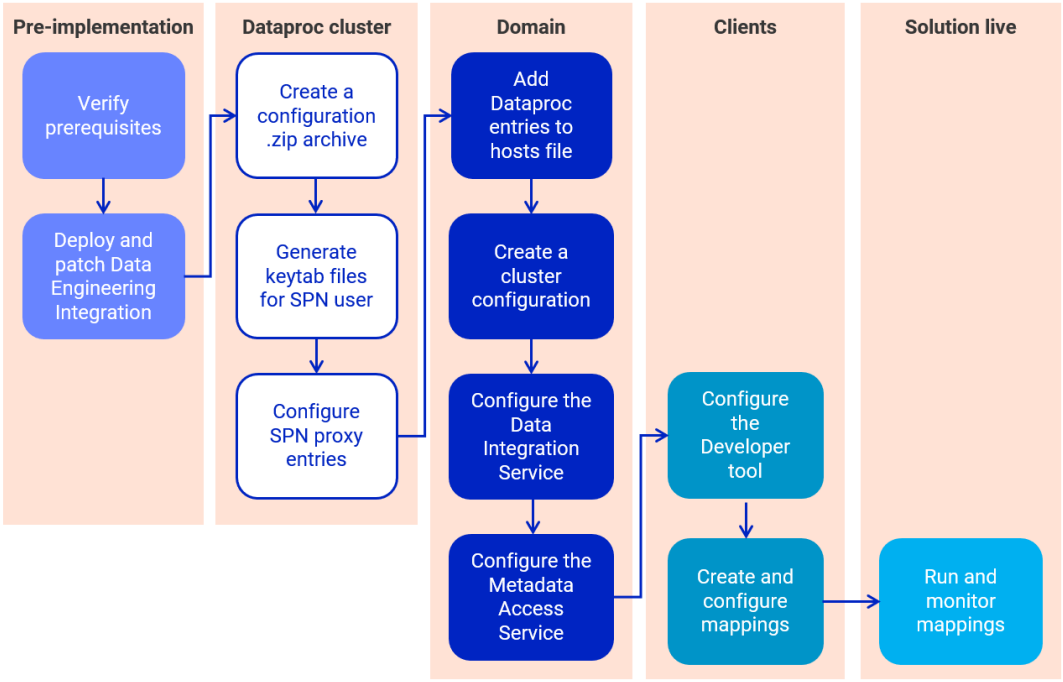
Google Dataproc is a lightweight implementation of Hadoop and Apache Spark on the Google cloud platform. When you integrate Informatica Data Engineering products with Dataproc, you configure an on-premises Informatica domain to run jobs on the Dataproc cloud cluster.

When configuration is complete, you can run mappings from the Developer tool in the Dataproc environment.

For information about supported security providers, see the topic "Supported Identity Providers" in the *Informatica Security Guide*.

For more information about product requirements and supported platforms, see the Product Availability Matrix on Informatica Network:
https://network.informatica.com/community/informatica-network/product-availability-matrices

The following image shows an overview of the integration process:



Pre-Implementation Tasks

Before you enable Data Engineering Integration to run jobs on Google Dataproc, perform the following pre-implementation tasks:

- Verify prerequisites.
- Create the Data Engineering Integration domain on-premises and apply the Dataproc implementation patch.

Verify Prerequisites

Verify the following prerequisites before you begin to configure Data Engineering Integration for use with Dataproc:

- You have a Google Dataproc account with permissions to create resources, rules and policies.
- You have purchased a Data Engineering Integration license.
- You have read the following information about support for Informatica's integration with Google Dataproc:
  - Data Engineering Integration Known Limitations in the *Informatica Release Notes*
  - Product Availability Matrix:
    https://network.informatica.com/community/informatica-network/product-availability-matrices
- You have verified that the Model Repository Service, Data Integration Service, and Metadata Access Service are running on the Informatica domain.
  See the *Informatica Installation for Data Engineering* guide.

*Configure *-site.xml Files for Google Dataproc*

The Hadoop administrator needs to configure *-site.xml file properties and restart impacted services before the Informatica administrator imports cluster information into the domain.

core-site.xml

Configure the following properties in the core-site.xml file:

fs.s3.enableServerSideEncryption

Enables server side encryption for S3 buckets. Required for SSE and SSE-KMS encryption.

Set to: TRUE

fs.s3a.access.key

The ID for the Blaze and Spark engines to connect to the Amazon S3 file system.

Set to your access key.

fs.s3a.secret.key

The password for the Blaze and Spark engines to connect to the Amazon S3 file system

Set to your access ID.

fs.s3a.server-side-encryption-algorithm

The server-side encryption algorithm for S3. Required for SSE and SSE-KMS encryption. Set to the encryption algorithm used.

hadoop.proxyuser.<proxy user>.groups

Defines the groups that the proxy user account can impersonate. On a secure cluster the <proxy user> is the Service Principal Name that corresponds to the cluster keytab file. On a non-secure cluster, the <proxy user> is the system user that runs the Informatica daemon.

Set to group names of impersonation users separated by commas. If less security is preferred, use the wildcard " * " to allow impersonation from any group.

Note: After you make changes to proxyuser properties, restart the credential service and other cluster services that use core-site configuration values.

hadoop.proxyuser.<proxy user>.hosts

Defines the host machines that a user account can impersonate. On a secure cluster the <proxy user> is the Service Principal Name that corresponds to the cluster keytab file. On a non-secure cluster, the <proxy user> is the system user that runs the Informatica daemon.

Set to a single host name or IP address, or set to a comma-separated list. If less security is preferred, use the wildcard " * " to allow impersonation from any host.

Note: After you make changes to proxyuser properties, restart the credential service and other cluster services that use core-site configuration values.

hadoop.proxyuser.hive.hosts

Comma-separated list of hosts that you want to allow the Hive user to impersonate on a non-secure cluster.

When `hive.server2.enable.doAs` is false, append a comma-separated list of Informatica server host names or IP address where the Data Integration Service is running. If less security is preferred, use the wildcard " * " to allow impersonation from any host.

Note: After you make changes to this property, restart the cluster services that use core-site configuration values.

io.compression.codecs

Enables compression on temporary staging tables.

Set to a comma-separated list of compression codec classes on the cluster.

hadoop.security.auth_to_local

Translates the principal names from the Active Directory and MIT realm into local names within the Hadoop cluster. Based on the Hadoop cluster used, you can set multiple rules.

Set to: RULE:[1:$1@$0](^.*@YOUR.REALM)s/^(.*)@YOUR.REALM\.COM$/$1/g

Set to: RULE:[2:$1@$0](^.*@YOUR.REALM\.$)s/^(.*)@YOUR.REALM\.COM$/$1/g

fair-scheduler.xml

Configure the following properties in the fair-scheduler.xml file:

allowPreemptionFrom

Enables preemption for the Fair Scheduler. The Blaze engine does not support preemption. If YARN preemption is enabled for the cluster, you need to disable it for the queue allocated to the Blaze engine.

Set to FALSE for the queue allocated to the Blaze engine.

For example:

```
<queue name="Blaze">
    <weight>1.0</weight>
    <allowPreemptionFrom>false</allowPreemptionFrom>
    <schedulingPolicy>fsp</schedulingPolicy>
    <aclSubmitApps>*</aclSubmitApps>
    <aclAdministerApps>*</aclAdministerApps>
</queue>
```

hbase-site.xml

Configure the following properties in the hbase-site.xml file:

zookeeper.znode.parent

Identifies HBase master and region servers.

Set to the relative path to the znode directory of HBase.

hdfs-site.xml

Configure the following properties in the hdfs-site.xml file:

dfs.encryption.key.provider.uri

The KeyProvider used to interact with encryption keys when reading and writing to an encryption zone. Required if sources or targets reside in the HDFS encrypted zone on Java KeyStore KMS-enabled Cloudera CDH cluster or a Ranger KMS-enabled Hortonworks HDP cluster.

Set to: kmf://http@xx11.xyz.com:16000/kms

dfs.namenode.rpc-bind-host

The actual address the Remote Procedure Call (RPC) server will bind to. If this optional address is set, it overrides the hostname portion of dfs.namenode.rpc-address. Enables the cluster to listen on private and public network interfaces, allowing remote access and datanode access. Required when you run mappings on a non-VPN Dataproc cluster.

Set to: 0.0.0.0 to enable the cluster to listen on private and public ports, allowing remote access and datanode access.

dfs.namenode.servicerpc-bind-host

The actual address the Remote Procedure Call (RPC) server will bind to. If this optional address is set, it overrides the hostname portion of dfs.namenode.rpc-address. Enables the cluster to listen on private and public network interfaces, allowing remote access and datanode access. Required when you run mappings on a non-VPN Dataproc cluster.

Set to: 0.0.0.0 to enable the cluster to listen on private and public ports, allowing remote access and datanode access.

dfs.namenode.http-bind-host

The actual address the Remote Procedure Call (RPC) server will bind to. If this optional address is set, it overrides the hostname portion of dfs.namenode.rpc-address. Enables the cluster to listen on private and public network interfaces, allowing remote access and datanode access. Required when you run mappings on a non-VPN Dataproc cluster.

Set to: 0.0.0.0 to enable the cluster to listen on private and public ports, allowing remote access and datanode access.

dfs.namenode.https-bind-host

The actual address the Remote Procedure Call (RPC) server will bind to. If this optional address is set, it overrides the hostname portion of dfs.namenode.rpc-address. Enables the cluster to listen on private and public network interfaces, allowing remote access and datanode access. Required when you run mappings on a non-VPN Dataproc cluster.

Set to: 0.0.0.0 to enable the cluster to listen on private and public ports, allowing remote access and datanode access.

hive-site.xml

Configure the following properties in the hive-site.xml file:

hive.async.log.enabled

Enables asynchronous logging. Required when you enable pre-task and post-task monitoring statistics on a Dataproc cluster.

Set to: FALSE

hive.cluster.delegation.token.store.class

The token store implementation. Required for HiveServer2 high availability and load balancing.

Set to: org.apache.hadoop.hive.thrift.DBTokenStore

hive.exec.dynamic.partition

Enables dynamic partitioned tables for Hive tables. Applicable for Hive versions 0.9 and earlier.

Set to: TRUE

hive.exec.dynamic.partition.mode

Allows all partitions to be dynamic. Required if you use Sqoop and define a DDL query to create or replace a partitioned Hive target at run time.

Set to: nonstrict

hive.server2.in.place.progress

Allows HiveServer2 to send progress bar update information. Takes effect only when you enable Tez. Required when you enable pre-task and post-task monitoring statistics on a Dataproc cluster.

Set to: TRUE

hive.server2.logging.operation.level

Hive Server2 logging level at the session level. Requires hive.server2.logging.operation.enabled to be set to TRUE. Required when you enable pre-task and post-task monitoring statistics on a Dataproc cluster.

Set to: EXECUTION

hive.server2.logging.operation.enabled

Enables logs to be saved. Required when you enable pre-task and post-task monitoring statistics on a Dataproc cluster.

Set to: TRUE

hive.server2.zookeeper.namespace

The value of the ZooKeeper namespace in the JDBC connection string. Required for HiveServer2 high availability.

Set to: `jdbc:hive2://<zookeeper_ensemble>/default;serviceDiscoveryMode=zooKeeper;zooKeeperNamespace=hiveserver2`

hive.zookeeper.quorum

Comma-separated list of ZooKeeper server host:ports in a cluster. The value of the ZooKeeper ensemble in the JDBC connection string. Required for HiveServer2 high availability.

Set to: `jdbc:hive2://<zookeeper_ensemble>/default;serviceDiscoveryMode=zooKeeper;`

mapred-site.xml

Configure the following properties in the mapred-site.xml file:

mapreduce.framework.name

The run-time framework to run MapReduce jobs. Values can be local, classic, or yarn. Required for Sqoop.

Set to: yarn

mapreduce.jobhistory.address

Location of the MapReduce JobHistory Server. The default port is 10020. Required for Sqoop.

Set to: <MapReduce JobHistory Server>:<port>

mapreduce.jobhistory.intermediate-done-dir

Directory where MapReduce jobs write history files. Required for Sqoop.

Set to: /mr-history/tmp

mapreduce.jobhistory.done-dir

Directory where the MapReduce JobHistory Server manages history files. Required for Sqoop.

Set to: /mr-history/done

mapreduce.jobhistory.principal

The Service Principal Name for the MapReduce JobHistory Server. Required for Sqoop.

Set to: mapred/_HOST@YOUR-REALM

mapreduce.jobhistory.webapp.address

Web address of the MapReduce JobHistory Server. The default value is 19888. Required for Sqoop.

Set to: <host>:<port>

yarn.app.mapreduce.am.staging-dir

The HDFS staging directory used while submitting jobs.

Set to the staging directory path.

tez-site.xml

Configure the following properties in the tez-site.xml file:

tez.am.tez-ui.history-url.template

Tez UI URL template for the application. The application manager uses this URL to redirect the user to the Tez UI. Required when you enable pre-task and post-task monitoring statistics on a Dataproc cluster.

Set value to:

```
_HISTORY_URL_BASE?%2F%23%2Ftez-app%2FAPPLICATION_ID
```

Note: The values of `_HISTORY_URL_BASE_` and `_APPLICATION_ID` are resolved at runtime. Do not edit the string to supply values.

tez.runtime.io.sort.mb

The sort buffer memory. Required when the output needs to be sorted for Blaze and Spark engines.

Set value to 270 MB.

tez.task.generate.counters.per.io

Enables pre-task and post-task monitoring statistics on an Amazon EMR or Dataproc cluster.

Set to: TRUE

yarn-site.xml

Configure the following properties in the yarn-site.xml file:

yarn.application.classpath

Required for dynamic resource allocation.

Add spark_shuffle.jar to the class path. The .jar file must contain the class "org.apache.spark.network.yarn.YarnShuffleService."

yarn.nodemanager.resource.memory-mb

The maximum RAM available for each container. Set the maximum memory on the cluster to increase resource memory available to the Blaze engine.

Set to 16 GB if value is less than 16 GB.

yarn.nodemanager.resource.cpu-vcores

The number of virtual cores for each container. Required for Blaze engine resource allocation.

Set to 10 if the value is less than 10.

yarn.scheduler.minimum-allocation-mb

The minimum RAM available for each container. Required for Blaze engine resource allocation.

Set to 6 GB if the value is less than 6 GB.

yarn.nodemanager.vmem-check-enabled

Disables virtual memory limits for containers. Required for the Blaze and Spark engines.

Set to: false

yarn.nodemanager.aux-services

Required for dynamic resource allocation for the Spark engine.

Add an entry for "spark_shuffle."

yarn.nodemanager.aux-services.spark_shuffle.class

Required for dynamic resource allocation for the Spark engine.

Set to: org.apache.spark.network.yarn.YarnShuffleService

yarn.resourcemanager.scheduler.class

Defines the YARN scheduler that the Data Integration Service uses to assign resources.

Set to: org.apache.hadoop.yarn.server.resourcemanager.scheduler

yarn.node-labels.enabled

Enables node labeling.

Set to: TRUE

yarn.node-labels.fs-store.root-dir

The HDFS location to update node label dynamically.

Set to: <hdfs://[Node name]:[Port]/[Path to store]/[Node labels]/>

Create a Cluster Configuration

Perform this task in the following situations:
- You are integrating for the first time. - You upgraded from any previous version.

A cluster configuration is an object in the domain that contains configuration information about the Hadoop cluster. The cluster configuration enables the Data Integration Service to push mapping logic to the Hadoop environment. Import configuration properties from the Hadoop cluster to create a cluster configuration.

The import process imports values from *-site.xml files into configuration sets based on the individual *-site.xml files. When you perform the import, the cluster configuration wizard can create Hadoop, HBase, HDFS, and Hive connection to access the Hadoop environment. If you choose to create the connections, the wizard also associates the cluster configuration with the connections.

Note: If you are integrating for the first time and you imported the cluster configuration when you ran the installer, you *must* re-create or refresh the cluster configuration.

Prepare the Archive File for Import from Google Dataproc

To provide an archive file for the Informatica administrator, ensure that you include all required site-*.xml files.

Create a .zip or .tar file that contains the following *-site.xml files:

- core-site.xml
- fair-scheduler.xml
- hbase-site.xml. Required only for access to HBase sources and targets.
- hdfs-site.xml
- hive-site.xml
- mapred-site.xml
- tez-site.xml
- yarn-site.xml

Give the Informatica administrator access to the archive file to import the cluster information into the domain.

Importing a Hadoop Cluster Configuration from a File

You can import properties from an archive file to create a cluster configuration.

Before you import from the cluster, you must get the archive file from the Hadoop administrator.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.
2. From the Actions menu, select **New** > **Cluster Configuration**.

   The **Cluster Configuration** wizard opens.
3. Configure the following properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The cluster Hadoop distribution type.
Method to import the cluster configuration	Choose **Import from file** to import properties from an archive file.
Create connections	Choose to create Hadoop, HDFS, Hive, and HBase connections. If you choose to create connections, the Cluster Configuration wizard associates the cluster configuration with each connection that it creates. The Hadoop connection contains default values for properties such as cluster environment variables, cluster path variables, and advanced properties. Based on the cluster environment and the functionality that you use, you can add to the default values or change the default values of these properties. For a list of Hadoop connection properties to configure, see "Configuring Hadoop Connection Properties" on page 10. If you do not choose to create connections, you must manually create them and associate the cluster configuration with them. **Important:** When the wizard creates the Hive connection, it populates the Metadata Connection String and the Data Access Connection String properties with the value from the hive.metastore.uris property. If the Hive metastore and HiveServer2 are running on different nodes, you must update the Metadata Connection String to point to the HiveServer2 host.

4. Click **Browse** to select a file. Select the file and click **Open**.
5. Click **Next** and verify the cluster configuration information on the summary page.

Configuring Hadoop Connection Properties

When you create a Hadoop connection, default values are assigned to cluster environment variables, cluster path properties, and advanced properties. You can add or edit values for these properties. You can also reset to default values.

You can configure the following Hadoop connection properties based on the cluster environment and functionality that you use:

- Cluster Environment Variables
- Cluster Library Path
- Common Advanced Properties
- Blaze Engine Advanced Properties

- Spark Engine Advanced Properties

Note: Informatica does not recommend changing these property values before you consult with third-party documentation, Informatica documentation, or Informatica Global Customer Support. If you change a value without knowledge of the property, you might experience performance degradation or other unexpected results.

To reset to default values, delete the property values. For example, if you delete the values of an edited Cluster Library Path property, the value resets to the default $DEFAULT_CLUSTER_LIBRARY_PATH.

Verify the Distribution Version

Verify that the Distribution version property is set to 2.0.

In Informatica version 10.5.1x and lower, Informatica creates the cluster configuration with the Distribution version property set to "1.4 (default)" by default. To integrate with Dataproc 2.x clusters, manually update the Distribution version property to 2.0.

1. In the Administrator tool, click **Connections.**
2. Expand the **Cluster Configurations** node in the Domain Navigator and select the Dataproc cluster configuration.
3. Set the Distribution Version property to `2.0`.
4. Save the changes and restart the domain.

Configure the Cluster for the Blaze Engine

To run mappings on the Blaze engine, the cluster requires the libncurses.so.5 library.

Perform this task in the following situations:
- You are integrating for the first time. - You upgraded from any previous version.

You can use the following command to install the 64-bit libncurses.so.5 library on all cluster nodes:

```
sudo apt-get install libncurses5
```

Configure Domain Settings

Perform this task in the following situations:
- You are integrating for the first time. - You upgraded from any previous version.

Perform the following steps to configure the Informatica domain to run mappings on the Dataproc cluster:

1. Copy the krb5.conf file from the cluster to the domain.
2. Enable access to Google Cloud Services.
3. Verify JDBC drivers to enable Sqoop connectivity.

Copy the Kerberos Configuration File

Copy the Kerberos configuration file Krb5.conf from the cluster to the following directories:

- <Informatica installation directory>/services/shared/security
- <Informatica installation directory>/java/jre/lib/security

Enable Access to Google Cloud Sources

You can use Google Cloud resources as mapping sources. You can enable access to Google Cloud storage by saving account credentials in a JSON file on each Data Integration Service machine.

1. Create service account credentials in JSON file format.

   Set permissions for the account to access the Google Cloud resources that act as data sources for mappings. To create the JSON credentials file, see the Google documentation.
2. Save the JSON file on the machine that hosts the Data Integration Service. Set full permissions to enable the Data Integration Service to access the file.
3. In the Administrator tool, select the Data Integration Service and edit the Environment Variables properties to point to the JSON file.

   Add the following key-value pair to the environment variables:

   Key: GOOGLE_APPLICATION_CREDENTIALS

   Value: `<path to the JSON file>`
4. Recycle the Data Integration Service.

Note: Perform these steps on each machine that hosts the Data Integration Service.

Verify JDBC Drivers for Sqoop Connectivity

Verify that you have the JDBC drivers to access JDBC-compliant databases in the Hadoop environment. You might need separate drivers for metadata import and for run-time processing.

Perform this task in the following situations:
- You are integrating for the first time.

You download drivers based on design-time and run-time requirements:

- **Design-time.** To import metadata, you can use the DataDirect drivers packaged with the Informatica installer if they are available. If they are not available, use any Type 4 JDBC driver that the database vendor recommends.
- **Run-time.** To run mappings, use any Type 4 JDBC driver that the database vendor recommends. Some distributions support other drivers to use Sqoop connectors. You cannot use the DataDirect drivers for run-time processing.

Verify Design-time Drivers

Use the DataDirect JDBC drivers packaged with the Informatica installer to import metadata from JDBC-compliant databases. If the DataDirect JDBC drivers are not available for a specific JDBC-compliant database, download the Type 4 JDBC driver associated with that database.

Copy the JDBC driver .jar files to the following location on the Developer tool machine:

```
<Informatica installation directory>\clients\externaljdbcjars
```

Verify Run-time Drivers

Verify run-time drivers for mappings that access JDBC-compliant databases in the Hadoop environment. Use any Type 4 JDBC driver that the database vendor recommends.

1. Download Type 4 JDBC drivers associated with the JCBC-compliant databases that you want to access.
2. To optimize the Sqoop mapping performance on the Spark engine while writing data to an HDFS complex file target of the Parquet format, download the following .jar files:
   - `parquet-hadoop-bundle-1.6.0.jar` from https://repo1.maven.org/maven2/com/twitter/parquet-hadoop-bundle/1.6.0/
   - `parquet-avro-1.6.0.jar` from https://repo1.maven.org/maven2/com/twitter/parquet-avro/1.6.0/
   - `parquet-column-1.5.0.jar` from https://repo1.maven.org/maven2/com/twitter/parquet-column/1.5.0/
3. Copy all of the .jar files to the following directory on the machine where the Data Integration Service runs:

   ```
   <Informatica installation directory>\externaljdbcjars
   ```

   Changes take effect after you recycle the Data Integration Service. At run time, the Data Integration Service copies the .jar files to the Hadoop distribution cache so that the .jar files are accessible to all nodes in the cluster.

Enable Access to Google Cloud Sources

You can use Google Cloud resources as mapping sources. You can enable access to Google Cloud storage by saving account credentials in a JSON file on each Data Integration Service machine.

1. Create service account credentials in JSON file format.

   Set permissions for the account to access the Google Cloud resources that act as data sources for mappings. To create the JSON credentials file, see Google Cloud documentation.
2. Save the JSON file on the machine that hosts the Data Integration Service. Set full permissions to enable the Data Integration Service to access the file.
3. In the Administrator tool, select the Data Integration Service and edit the Environment Variables properties to point to the JSON file.

   Add the following key-value pair to the environment variables:

   Key: GOOGLE_APPLICATION_CREDENTIALS

   Value: `<path to the JSON file>`
4. Recycle the Data Integration Service.

Note: Perform these steps on each machine that hosts the Data Integration Service.

Configure the Developer Tool

Perform these steps on each machine that hosts the Developer tool.

Perform this task in the following situations:
- You are integrating for the first time. - You upgraded from any previous version.

Edit the hosts File for Google Dataproc

To enable Informatica to access the Dataproc cluster, edit the `/etc/hosts` file on the domain host.

Enter the IP address, DNS name, and DNS short name for each data node on the cluster.

For example:

```
10.20.30.40 dataprocABC-m.c.MyUsernameINFA54321.internal dataprocABC-m
10.20.30.44 dataprocABC-w-0.c.MyUsernameINFA54321.internal dataprocABC-w-0
10.20.30.43 dataprocABC-w-1.c.MyUsernameINFA54321.internal dataprocABC-w-1
10.20.30.42 dataprocABC-w-2.c.MyUsernameINFA54321.internal dataprocABC-w-2
```

Edit the hosts file with cluster node information on each machine that hosts the Data Integration Service.

Configure developerCore.ini

Edit the developerCore.ini file to import complex files.

Edit the developerCore.ini file on each machine that hosts the Developer tool.

When you import a complex file, such as Avro or Parquet, the imported object includes metadata associated with the distribution in the Hadoop environment. If the file resides on the Developer tool machine, the import process picks up the distribution information from the developerCore.ini file. You must edit the developerCore.ini file to point to the distribution directory on the Developer tool machine.

You can find the developerCore.ini file in the following directory:

```
<Informatica installation directory>\clients\DeveloperClient
```

Add the following property:

```
-DINFA_HADOOP_DIST_DIR=hadoop\<distribution>_<version>
```

The change takes effect when you restart the Developer tool.

Configure the Developer Tool for Kerberos

To import metadata from Hive, HBase, and complex file sources, import configuration files from the Kerberos-enabled cluster, and generate the Kerberos credentials file on the Developer tool machine.

Import configuration files

The Hadoop cluster uses a set of XML files named *-site.xml to store configuration settings. The domain uses the same set of files to create the cluster configuration object.

To enable you to import metadata from the cluster, import the *-site.xml files to each Developer tool machine:

1. Log in to the Administrator tool and navigate to **Connections** > **Cluster Configuration** > **CCO**. Locate the cluster configuration associated with the Hadoop cluster.
2. Extract the *-site.xml files in the cluster configuration, inlcuding sensitive properties, to the following directory on the Developer tool machine: `<Informatica installation directory>\clients\DeveloperClient\hadoop\<Hadoop distribution>\conf`
   For more information about sensitive properties, see the "Active Properties" topic in the *Data Engineering Administrator Guide*.

Note: If you refresh the cluster configuration, repeat these steps.

Generate the Kerberos credentials file

1. Copy the krb5.conf file from `<Developer tool installation directory>/services/shared/security` to `C:/Windows`.
2. Rename krb5.conf to krb5.ini.

3. In the krb5.ini file, verify the value of the forwardable option to determine how to use the kinit command. If `forwardable=true`, run the command with the `-f` option. Otherwise, run the command without the -f option.
4. To generate the Kerberos credentials file, run the kinit command from the following location: `<Developer tool installation directory>/clients/java/bin/kinit.exe`
   For example, you might run the following command: `kinit joe/domain12345@MY-REALM`

Configure Mappings to Run on Dataproc

To run mappings on the Dataproc cluster, configure mappings with the following properties:

1. In the **Parameters** section, create a parameter with the values shown in the following table:

Name	Type	Precision	Scale	Default Value
pushdownConnection	Connection	(blank)	(blank)	Use the dropdown control to select the Hadoop connection that the cluster configuration created.

2. In the **Run-Time** section, choose the following values:
   - Under Validation Environments, select Spark.
   - For Execution Environment, select Hadoop.
   - Under the Hadoop section, for the Connection property, click the right side of the Value cell to see the dropdown menu, and choose the pushdownConnection parameter.

Monitoring Data Engineering Jobs

To monitor the status of jobs on the Dataproc cluster, access the Hadoop Resource Manager.

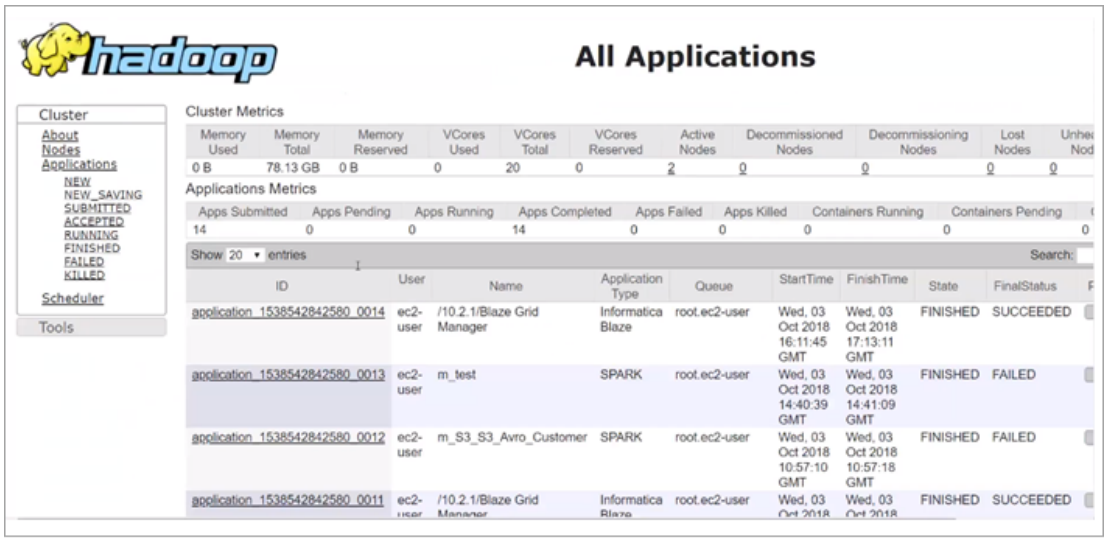
To access the Hadoop Resource Manager, browse to the following URL:

```
http://<Master_node_name>:8088
```

where the Master_node_name is the node name or IP address of the Dataproc master node. For example:

```
http://dataproc123-m:8088
```

The following image shows the Hadoop resource manager:



Troubleshooting Mapping Failures on the Dataproc Cluster

This section provides information on troubleshooting common error messages and limitations that you might encounter when you run mappings on a Dataproc cluster.

When you run mappings on a Dataproc cluster to read or write data to SerDe-backed Hive tables, the mapping fails with the java.lang.ClassNotFoundException: org.apache.hive.hcatalog.data.JsonSerDe error.

Add the following property in the `hive-site.xml` on all the nodes of the Dataproc cluster and restart the Hive server:

```
<property>
  <name>hive.aux.jars.path</name>
  <value>file:///usr/lib/hive/lib/hive-hcatalog-core-<version>.jar</value>
</property>
```

When you run mappings on a non-VPN Dataproc cluster, the mappings fail.

Configure the following properties in the `hdfs-site.xml` on all the nodes of the Dataproc cluster:

```
<property>
  <name>dfs.namenode.rpc-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.namenode.servicerpc-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.namenode.http-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.namenode.https-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.client.use.datanode.hostname</name>
  <value>true</value>
</property>
```

When I run a mapping with a Hive source or a Hive target on a different cluster, the Data Integration Service fails to push the mapping to Hadoop with the following error: `Failed to execute query [exec0_query_6] with error code [10], error message [FAILED: Error in semantic analysis: Line 1:181 Table not found customer_eur], and SQL state [42000]].`

When you run a mapping in a Hadoop environment, the Hive connection selected for the Hive source or Hive target, and the mapping must be on the same Hive metastore.

When I run a mapping with SQL overrides concurrently, the mapping hangs.

There are not enough available resources because the cluster is being shared across different engines.

Use different YARN scheduler queues for the Blaze and Spark engines to allow HiveServer2 to run SQL overrides through these engines.

When I configure a mapping to create a partitioned Hive table, the mapping fails with the error "Need to specify partition columns because the destination table is partitioned."

This issue happens because of internal Informatica requirements for a query that is designed to create a Hive partitioned table. For details and a workaround, see Knowledge Base article 516266.

Time stamp data that is precise to the nanosecond is truncated when a mapping runs on the Spark engine

Spark stores time stamp data to a precision of 1 microsecond (1us) and does not support nanosecond precision. When a mapping that runs on the Spark engine reads datetime data that has nanosecond

precision, the data is truncated to the microsecond. For example, `2015-01-02 00:00:00.000456789` is truncated to `2015-01-02 00:00:00.000456`.

The Blaze engine supports nanosecond precision.

Author